Below is a **complete developer note** you can keep in your repository (for example: docs/aws_deployment_codify.md).
It covers the **full AWS deployment workflow you just did** for your **Django + Gunicorn + Nginx + Domain + SSL** setup.

I wrote it like **professional project documentation**.

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**Codify Backend Deployment Guide (AWS EC2)**

Complete production deployment documentation for **Django + Gunicorn + Nginx + Domain + SSL**.

Server used:

- **AWS EC2 Ubuntu 22.04**
- **Python 3.12**
- **Gunicorn + UvicornWorker**
- **Nginx reverse proxy**
- **Domain:** backend.jitcoder.in

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**1. Server Provisioning**

Create EC2 instance.

Recommended configuration:

| Item | Value |
|---|---|
| Instance | t2.micro (Free tier) |
| OS | Ubuntu 22.04 LTS |
| Storage | 20GB |
| Security group | Allow 22, 80, 443 |

Ports:

22 SSH

80 HTTP

443 HTTPS

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**2. Connect to Server**

From local machine:

```
ssh -i "jitcoderDocker.pem" ubuntu@EC2_PUBLIC_IP
```

Example:

ssh -i "jitcoderDocker.pem" ubuntu@3.230.159.239

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**3. Update Server**

sudo apt update

sudo apt upgrade -y

Install required packages:

sudo apt install python3-pip python3-venv nginx redis-server git -y

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**4. Clone Project**

cd /home/ubuntu

git clone https://github.com/Dasabinash150/codify.git

cd codify

Project path:

/home/ubuntu/codify/codify

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**5. Create Virtual Environment**

cd /home/ubuntu/codify/codify

python3 -m venv venv

source venv/bin/activate

---

**6. Install Dependencies**

pip install -r requirements.txt

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**7. Environment Configuration**

Create production env file.

/home/ubuntu/codify/codify/.env.production

Example:

DEBUG=False

SECRET_KEY=your_secret_key

ALLOWED_HOSTS=backend.jitcoder.in,3.230.159.239,localhost,127.0.0.1

CSRF_TRUSTED_ORIGINS=https://backend.jitcoder.in

CORS_ALLOWED_ORIGINS=https://backend.jitcoder.in

DATABASE_URL=sqlite:///db.sqlite3

REDIS_URL=redis://127.0.0.1:6379

---

**8. Run Django Setup**

Activate environment:

source venv/bin/activate

export ENVIRONMENT=production

Run migrations:

python manage.py migrate

Collect static files:

python manage.py collectstatic

Create admin user:

python manage.py createsuperuser

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**9. Test Django Server**

gunicorn codify.asgi:application \

-k uvicorn_worker.UvicornWorker \

--bind 0.0.0.0:8000

Test:

http://EC2_IP:8000

Stop with:

CTRL + C

---

**10. Configure Gunicorn Service**

Create systemd service.

sudo nano /etc/systemd/system/gunicorn.service

Example:

```
[Unit]
Description=Gunicorn daemon for Codify ASGI
After=network.target

[Service]
User=ubuntu
Group=www-data
WorkingDirectory=/home/ubuntu/codify/codify
Environment="PATH=/home/ubuntu/codify/codify/venv/bin"
Environment="ENVIRONMENT=production"

ExecStart=/home/ubuntu/codify/codify/venv/bin/gunicorn \
codify.asgi:application \
-k uvicorn_worker.UvicornWorker \
--workers 3 \
--bind unix:/home/ubuntu/codify/codify/gunicorn.sock

[Install]
WantedBy=multi-user.target
```

Start service:

```
sudo systemctl daemon-reload
sudo systemctl start gunicorn
sudo systemctl enable gunicorn
```

Check:

```
sudo systemctl status gunicorn
```

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**11. Configure Nginx**

Create config:

sudo nano /etc/nginx/sites-available/codify

Example:

```
server {
    listen 80;
    server_name backend.jitcoder.in;

    location /static/ {
        alias /home/ubuntu/codify/codify/staticfiles/;
    }

    location /media/ {
        alias /home/ubuntu/codify/codify/media/;
    }

    location /ws/ {
        proxy_pass http://unix:/home/ubuntu/codify/codify/gunicorn.sock;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "upgrade";
        proxy_set_header Host $host;
    }

    location / {
        include proxy_params;
        proxy_pass http://unix:/home/ubuntu/codify/codify/gunicorn.sock;
    }
}
```

Enable config:

sudo ln -s /etc/nginx/sites-available/codify /etc/nginx/sites-enabled

Test config:

sudo nginx -t

Restart:

sudo systemctl restart nginx

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**12. Domain Configuration**

In DNS provider create record:

| Type | Name | Value |
|---|---|---|
| A | backend | EC2_PUBLIC_IP |

Example:

backend.jitcoder.in → 3.230.159.239

Wait for DNS propagation.

Test:

ping backend.jitcoder.in

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**13. Install SSL Certificate**

Install certbot:

sudo apt install certbot python3-certbot-nginx -y

Run:

sudo certbot --nginx -d backend.jitcoder.in

Choose:

Redirect HTTP to HTTPS

Final URL:

https://backend.jitcoder.in

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**14. Restart Services**

sudo systemctl restart gunicorn

sudo systemctl restart nginx

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**15. Verify Deployment**

Test endpoints:

https://backend.jitcoder.in

https://backend.jitcoder.in/admin

https://backend.jitcoder.in/api/contests/

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**16. Admin Permission Fix**

If admin dashboard shows:

You don't have permission to view or edit anything

Fix user permission.

Open shell:

```
python manage.py shell
```

Run:

```
from account.models import User

u = User.objects.get(email="admin@jitcoder.in")
u.is_superuser = True
u.save()
```

Restart gunicorn:

```
sudo systemctl restart gunicorn
```

---

**17. Useful Commands**

Check services:

```
sudo systemctl status gunicorn
sudo systemctl status nginx
sudo systemctl status redis-server
```

Restart:

```
sudo systemctl restart gunicorn
sudo systemctl restart nginx
```

View logs:

```
sudo journalctl -u gunicorn -n 100
sudo tail -f /var/log/nginx/error.log
```

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**18. Production Architecture**

Frontend (Vercel)

↓

https://backend.jitcoder.in

↓

Nginx Reverse Proxy

↓

Gunicorn + Django + Channels

↓

Redis

↓

Database

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**19. Next Production Improvements**

Recommended improvements:

1. PostgreSQL instead of SQLite
2. Redis queue for async jobs
3. Celery worker
4. Automatic DB backups
5. Monitoring (Prometheus / Sentry)
6. CDN for static files
7. Rate limiting
8. Docker deployment

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**20. Project Folder Structure**

```
codify/
 ├– account/
 ├– contest/
 ├– judge/
 ├– codify/
 |   ├– settings.py
 |   ├– asgi.py
 |   ├– wsgi.py
```

```
├── staticfiles/
├── media/
├── venv/
├── .env.production
├── manage.py
```

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**Deployment Result**

Backend successfully deployed:

https://backend.jitcoder.in

Admin panel:

https://backend.jitcoder.in/admin

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If you want, I can also give you a **much more advanced Codify production architecture used by real coding platforms (CodeChef / HackerRank style)** including:

- autoscaling workers
- code execution sandbox
- judge architecture
- job queues
- submission workers

That will help your **Codify platform scale to thousands of users**.